

Urban Taxi/Rideshare Bottlenecks

DPA PROJECT

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Screenshots start from page 2.

Project description:

Standard tools process data sequentially. Because the real-world NYC Taxi dataset is massive, processing it sequentially on a single machine would create a massive bottleneck.

This is a true Hadoop project because it utilizes HDFS for distributed storage and MapReduce for distributed parallel processing. Instead of reading one massive file, HDFS splits the data into blocks and distributes them across the nodes. Following the core Hadoop principle, the MapReduce framework then moves the computation (the Mapper tasks) to the nodes where the data is stored locally, which minimizes network traffic. We used standard `hadoop fs -mkdir`, `-put`, and `-cat` commands to manage this data flow just like we practiced in the labs.

1. The Mappers (Parallel Extraction)

Hadoop splits the massive CSV file and assigns chunks of it to multiple Mappers running in parallel. Our TaxiMapper reads the data line-by-line. It extracts the Hour from the pickup time and calculates the trip duration. The Output: It emits Key/Value pairs, where the Key is the Hour and the Value is the duration in minutes (e.g., ["08:00", 30.0]).

2. Shuffle and Sort (The Magic)

As the Map tasks finish, Hadoop automatically begins the Shuffle and Sort phase. This phase consolidates the intermediate data and guarantees that all trip durations associated with the exact same Hour key (e.g., "08:00") are routed to the exact same Reducer.

3. The Reducer (Aggregation)

Our AverageReducer receives the sorted intermediate keys and a list of all their associated values. For example, it receives: ["08:00", (30.0, 45.0, 15.0...)]. The Reducer loops through this list, sums the durations, divides by the total count to find the average, and writes the final Key/Value pair to HDFS.

We considered using a Combiner to optimize network traffic, but they can only be used if the operation is associative and commutative. Because our reducer calculates an Average (which is not associative), it cannot be used as a Combiner.

Screenshots (demo)

In the following pages you can see screenshots from our project, similar to a demo.

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We start Hadoop and check with the JPS command if Hadoop components run ok. These commands are runned in the terminal, in Linux.

```
pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ start-dfs.sh
start-yarn.sh
● pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ jps
19010 DataNode
19634 NodeManager
19235 SecondaryNameNode
18840 NameNode
19501 ResourceManager
20223 Jps
```

After this we can set the input. The output is already created (since we previously used Hadoop in DPA labs). Because we are using a VM and folders are situated in temp, we use also other Bash command to set up the Hadoop for the demo, like:

```
hdfs dfs -rm -r /real_output
stop-all.sh
hdfs namenode -format
hdfs dfs -mkdir -p /real_input
hdfs dfs -put real_taxi_data.csv /real_input/
```

In our case the setup is ready, so in the next Screenshoot you can see the workflow.

```
● pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hdfs dfs -mkdir -p /real_input
2026-05-12 09:59:31,066 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
● pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hdfs dfs -put real_taxi_data.csv /real_input/
2026-05-12 09:59:45,337 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
```

Since we are ready to run the “demo”, we will use Hadoop to process the data.

I have loaded the 10-million row NYC Taxi dataset into HDFS. I am now going to run my compiled MapReduce JAR file to process it. See next Screenshoot.

```

* pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real_input /real_output
2026-05-12 10:01:29,997 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-05-12 10:01:30,497 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-05-12 10:01:30,566 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-05-12 10:01:30,566 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-05-12 10:01:30,698 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute you
r application with ToolRunner to remedy this.
2026-05-12 10:01:30,837 INFO input.FileInputFormat: Total input files to process : 1
2026-05-12 10:01:30,882 INFO mapreduce.JobSubmitter: number of splits:13
2026-05-12 10:01:31,067 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local789236497_0001
2026-05-12 10:01:31,067 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-12 10:01:31,201 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-05-12 10:01:31,202 INFO mapreduce.Job: Running job: job_local789236497_0001
2026-05-12 10:01:31,204 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-05-12 10:01:31,212 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:01:31,213 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:01:31,213 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:01:31,214 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2026-05-12 10:01:31,263 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-05-12 10:01:31,263 INFO mapred.LocalJobRunner: Starting task: attempt_local789236497_0001_m_000000_0
2026-05-12 10:01:31,286 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:01:31,286 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:01:31,286 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:01:31,302 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:01:31,307 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:0+134217728
2026-05-12 10:01:31,380 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:01:31,381 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:01:31,381 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:01:31,381 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:01:31,381 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:01:31,386 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:01:32,207 INFO mapreduce.Job: Job job_local789236497_0001 running in uber mode : false
2026-05-12 10:01:32,208 INFO mapreduce.Job: map 0% reduce 0%
2026-05-12 10:01:34,365 INFO mapred.LocalJobRunner:
2026-05-12 10:01:34,365 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:01:34,365 INFO mapred.MapTask: Spilling map output
2026-05-12 10:01:34,366 INFO mapred.MapTask: bufstart = 0; bufend = 11970392; bufvoid = 104857600
2026-05-12 10:01:34,366 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22794288(91177152); length = 3420109/6553600
2026-05-12 10:01:34,667 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:01:34,681 INFO mapred.Task: Task:attempt_local789236497_0001_m_000000_0 is done. And is in the process of committing

```

This gives us the following result:

(Below is a list of Screenshoots that represent the terminal, when we run the program)

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real_input /real_output
2026-05-12 10:01:34,681 INFO mapred.Task: Task:attempt_local789236497_0001_m_000000_0 is done. And is in the process of committing
2026-05-12 10:01:34,686 INFO mapred.LocalJobRunner: map
2026-05-12 10:01:34,686 INFO mapred.Task: Task 'attempt_local789236497_0001_m_000000_0' done.
2026-05-12 10:01:34,692 INFO mapred.Task: Final Counters for attempt_local789236497_0001_m_000000_0: Counters: 23
File System Counters
  FILE: Number of bytes read=5401
  FILE: Number of bytes written=14390981
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=134221824
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=5
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=1
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=857810
  Map output records=855028
  Map output bytes=11970392
  Map output materialized bytes=13680454
  Input split bytes=116
  Combine input records=0
  Spilled Records=855028
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=69
  Total committed heap usage (bytes)=528482304
File Input Format Counters
  Bytes Read=134221824
2026-05-12 10:01:34,692 INFO mapred.LocalJobRunner: Finishing task: attempt_local789236497_0001_m_000000_0
2026-05-12 10:01:34,693 INFO mapred.LocalJobRunner: Starting task: attempt_local789236497_0001_m_000001_0
2026-05-12 10:01:34,694 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:01:34,694 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:01:34,694 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:01:34,695 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:01:34,696 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:134217728+134217728
2026-05-12 10:01:34,713 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:01:34,714 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:01:34,714 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:01:34,714 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:01:34,714 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:01:34,714 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:01:35,211 INFO mapreduce.Job: map 100% reduce 0%
2026-05-12 10:01:37,416 INFO mapred.LocalJobRunner:
2026-05-12 10:01:37,416 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:01:37,416 INFO mapred.MapTask: Spilling map output
2026-05-12 10:01:37,416 INFO mapred.MapTask: bufstart = 0; bufend = 12037732; bufvoid = 104857600
2026-05-12 10:01:37,416 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22775048(91100192); length = 3439349/6553600
2026-05-12 10:01:37,650 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:01:37,656 INFO mapred.Task: Task:attempt_local789236497_0001_m_000001_0 is done. And is in the process of committing
2026-05-12 10:01:37,660 INFO mapred.LocalJobRunner: map
2026-05-12 10:01:37,660 INFO mapred.Task: Task 'attempt_local789236497_0001_m_000001_0' done.

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
2026-05-12 10:01:37,660 INFO mapred.Task: Task 'attempt_local789236497_0001_m_000001_0' done.
2026-05-12 10:01:37,660 INFO mapred.Task: Final Counters for attempt_local789236497_0001_m_000001_0: Counters: 23
File System Counters
  FILE: Number of bytes read=6936
  FILE: Number of bytes written=28148427
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=268443648
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=7
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=1
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=861690
  Map output records=859838
  Map output bytes=12037732
  Map output materialized bytes=13757414
  Input split bytes=116
  Combine input records=0
  Spilled Records=859838
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=22
  Total committed heap usage (bytes)=529530880
File Input Format Counters
  Bytes Read=134221824
2026-05-12 10:01:37,661 INFO mapred.LocalJobRunner: Finishing task: attempt_local789236497_0001_m_000001_0
2026-05-12 10:01:37,661 INFO mapred.LocalJobRunner: Starting task: attempt_local789236497_0001_m_000002_0
2026-05-12 10:01:37,662 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:01:37,662 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:01:37,662 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:01:37,662 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:01:37,664 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:268435456+134217728
2026-05-12 10:01:37,688 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:01:37,688 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:01:37,688 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:01:37,688 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:01:37,688 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:01:37,689 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:01:40,453 INFO mapred.LocalJobRunner:
2026-05-12 10:01:40,453 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:01:40,453 INFO mapred.MapTask: Spilling map output
2026-05-12 10:01:40,453 INFO mapred.MapTask: bufstart = 0; bufend = 11945668; bufvoid = 104857600
2026-05-12 10:01:40,454 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22801352(91205408); length = 3413045/6553600
2026-05-12 10:01:40,692 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:01:40,697 INFO mapred.Task: Task:attempt_local789236497_0001_m_000002_0 is done. And is in the process of committing
2026-05-12 10:01:40,703 INFO mapred.LocalJobRunner: map
2026-05-12 10:01:40,704 INFO mapred.Task: Task 'attempt_local789236497_0001_m_000002_0' done.
2026-05-12 10:01:40,704 INFO mapred.Task: Final Counters for attempt_local789236497_0001_m_000002_0: Counters: 23
File System Counters
  FILE: Number of bytes read=8471

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
  FILE: Number of bytes read=8471
  FILE: Number of bytes written=41800657
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=402665472
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=9
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=1
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=855202
  Map output records=853262
  Map output bytes=11945668
  Map output materialized bytes=13652198
  Input split bytes=116
  Combine input records=0
  Spilled Records=853262
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=20
  Total committed heap usage (bytes)=529530880
File Input Format Counters
  Bytes Read=134221824
2026-05-12 10:01:40,704 INFO mapred.LocalJobRunner: Finishing task: attempt_local789236497_0001_m_000002_0
2026-05-12 10:01:40,704 INFO mapred.LocalJobRunner: Starting task: attempt_local789236497_0001_m_000003_0
2026-05-12 10:01:40,706 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:01:40,706 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:01:40,706 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:01:40,706 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:01:40,708 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:402653184+134217728
2026-05-12 10:01:40,724 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:01:40,725 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:01:40,725 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:01:40,725 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:01:40,725 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:01:40,725 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:01:43,338 INFO mapred.LocalJobRunner:
2026-05-12 10:01:43,338 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:01:43,338 INFO mapred.MapTask: Spilling map output
2026-05-12 10:01:43,338 INFO mapred.MapTask: bufstart = 0; bufend = 11983916; bufvoid = 104857600
2026-05-12 10:01:43,338 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22790424(91161696); length = 3423973/6553600
2026-05-12 10:01:43,593 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:01:43,598 INFO mapred.Task: Task:attempt_local789236497_0001_m_000003_0 is done. And is in the process of committing
2026-05-12 10:01:43,601 INFO mapred.LocalJobRunner: map
2026-05-12 10:01:43,601 INFO mapred.Task: Task 'attempt_local789236497_0001_m_000003_0' done.
2026-05-12 10:01:43,602 INFO mapred.Task: Final Counters for attempt_local789236497_0001_m_000003_0: Counters: 23
File System Counters
  FILE: Number of bytes read=10006
  FILE: Number of bytes written=55496599
  FILE: Number of read operations=0
  FILE: Number of large read operations=0

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=536887296
HDFS: Number of bytes written=0
HDFS: Number of read operations=11
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0

Map-Reduce Framework
  Map input records=857981
  Map output records=855994
  Map output bytes=11983916
  Map output materialized bytes=13695910
  Input split bytes=116
  Combine input records=0
  Spilled Records=855994
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=17
  Total committed heap usage (bytes)=530579456

File Input Format Counters
  Bytes Read=134221824
2026-05-12 10:01:43,602 INFO mapred.LocalJobRunner: Finishing task: attempt local789236497_0001_m_000003_0
2026-05-12 10:01:43,602 INFO mapred.LocalJobRunner: Starting task: attempt local789236497_0001_m_000004_0
2026-05-12 10:01:43,603 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:01:43,603 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:01:43,603 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:01:43,603 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:01:43,605 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:536870912+134217728
2026-05-12 10:01:43,620 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:01:43,620 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:01:43,620 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:01:43,620 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:01:43,620 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:01:43,621 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:01:46,208 INFO mapred.LocalJobRunner:
2026-05-12 10:01:46,208 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:01:46,208 INFO mapred.MapTask: Spilling map output
2026-05-12 10:01:46,208 INFO mapred.MapTask: bufstart = 0; bufend = 11956000; bufvoid = 104857600
2026-05-12 10:01:46,208 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22798400(91193600); length = 3415997/6553600
2026-05-12 10:01:46,218 INFO mapreduce.Job: map 31% reduce 0%
2026-05-12 10:01:46,442 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:01:46,447 INFO mapred.Task: Task:attempt local789236497_0001_m_000004_0 is done. And is in the process of committing
2026-05-12 10:01:46,451 INFO mapred.LocalJobRunner: map
2026-05-12 10:01:46,451 INFO mapred.Task: Task 'attempt local789236497_0001_m_000004_0' done.
2026-05-12 10:01:46,452 INFO mapred.Task: Final Counters for attempt local789236497_0001_m_000004_0: Counters: 23
File System Counters
  FILE: Number of bytes read=11541
  FILE: Number of bytes written=69160637
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=671109120

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
HDFS: Number of bytes read=671109120
HDFS: Number of bytes written=0
HDFS: Number of read operations=13
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0

Map-Reduce Framework
  Map input records=856064
  Map output records=854000
  Map output bytes=11956000
  Map output materialized bytes=13664006
  Input split bytes=116
  Combine input records=0
  Spilled Records=854000
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=17
  Total committed heap usage (bytes)=531628032

File Input Format Counters
  Bytes Read=134221824
2026-05-12 10:01:46,452 INFO mapred.LocalJobRunner: Finishing task: attempt local789236497_0001_m_000004_0
2026-05-12 10:01:46,452 INFO mapred.LocalJobRunner: Starting task: attempt local789236497_0001_m_000005_0
2026-05-12 10:01:46,453 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:01:46,453 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:01:46,453 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:01:46,453 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:01:46,455 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:671088640+134217728
2026-05-12 10:01:46,508 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:01:46,508 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:01:46,509 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:01:46,509 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:01:46,509 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:01:46,509 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:01:47,219 INFO mapreduce.Job: map 100% reduce 0%
2026-05-12 10:01:49,147 INFO mapred.LocalJobRunner:
2026-05-12 10:01:49,147 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:01:49,147 INFO mapred.MapTask: Spilling map output
2026-05-12 10:01:49,147 INFO mapred.MapTask: bufstart = 0; bufend = 11955538; bufvoid = 104857600
2026-05-12 10:01:49,147 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22798532(91194128); length = 3415865/6553600
2026-05-12 10:01:49,220 INFO mapreduce.Job: map 38% reduce 0%
2026-05-12 10:01:49,378 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:01:49,383 INFO mapred.Task: Task:attempt local789236497_0001_m_000005_0 is done. And is in the process of committing
2026-05-12 10:01:49,386 INFO mapred.LocalJobRunner: map
2026-05-12 10:01:49,386 INFO mapred.Task: Task 'attempt local789236497_0001_m_000005_0' done.
2026-05-12 10:01:49,386 INFO mapred.Task: Final Counters for attempt local789236497_0001_m_000005_0: Counters: 23
File System Counters
  FILE: Number of bytes read=12564
  FILE: Number of bytes written=82824147
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=805330944
  HDFS: Number of bytes written=0

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
HDFS: Number of bytes read=805330944
HDFS: Number of bytes written=0
HDFS: Number of read operations=15
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=856026
  Map output records=853967
  Map output bytes=11955538
  Map output materialized bytes=13663478
  Input split bytes=116
  Combine input records=0
  Spilled Records=853967
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=19
  Total committed heap usage (bytes)=637534208
File Input Format Counters
  Bytes Read=134221824
2026-05-12 10:01:49,387 INFO mapred.LocalJobRunner: Finishing task: attempt local789236497_0001_m_000005_0
2026-05-12 10:01:49,387 INFO mapred.LocalJobRunner: Starting task: attempt local789236497_0001_m_000006_0
2026-05-12 10:01:49,388 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:01:49,388 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:01:49,388 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:01:49,389 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:01:49,390 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:805306368+134217728
2026-05-12 10:01:49,408 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:01:49,408 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:01:49,408 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:01:49,408 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:01:49,408 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:01:49,409 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:01:50,221 INFO mapreduce.Job: map 100% reduce 0%
2026-05-12 10:01:52,079 INFO mapred.LocalJobRunner:
2026-05-12 10:01:52,079 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:01:52,079 INFO mapred.MapTask: Spilling map output
2026-05-12 10:01:52,079 INFO mapred.MapTask: bufstart = 0; bufend = 11957834; bufvoid = 104857600
2026-05-12 10:01:52,079 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22797876(91191504); length = 3416521/6553600
2026-05-12 10:01:52,222 INFO mapreduce.Job: map 46% reduce 0%
2026-05-12 10:01:52,317 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:01:52,322 INFO mapred.Task: Task:attempt local789236497_0001_m_000006_0 is done. And is in the process of committing
2026-05-12 10:01:52,325 INFO mapred.LocalJobRunner: map
2026-05-12 10:01:52,325 INFO mapred.Task: Task 'attempt local789236497_0001_m_000006_0' done.
2026-05-12 10:01:52,326 INFO mapred.Task: Final Counters for attempt_local789236497_0001_m_000006_0: Counters: 23
File System Counters
  FILE: Number of bytes read=13587
  FILE: Number of bytes written=96490281
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=939552768
  HDFS: Number of bytes written=0

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
HDFS: Number of bytes written=0
HDFS: Number of read operations=17
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=856198
  Map output records=854131
  Map output bytes=11957834
  Map output materialized bytes=13666102
  Input split bytes=116
  Combine input records=0
  Spilled Records=854131
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=14
  Total committed heap usage (bytes)=637534208
File Input Format Counters
  Bytes Read=134221824
2026-05-12 10:01:52,326 INFO mapred.LocalJobRunner: Finishing task: attempt local789236497_0001_m_000006_0
2026-05-12 10:01:52,326 INFO mapred.LocalJobRunner: Starting task: attempt local789236497_0001_m_000007_0
2026-05-12 10:01:52,327 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:01:52,327 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:01:52,327 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:01:52,327 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:01:52,328 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:939524096+134217728
2026-05-12 10:01:52,345 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:01:52,345 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:01:52,345 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:01:52,345 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:01:52,345 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:01:52,346 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:01:53,223 INFO mapreduce.Job: map 100% reduce 0%
2026-05-12 10:01:54,968 INFO mapred.LocalJobRunner:
2026-05-12 10:01:54,968 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:01:54,968 INFO mapred.MapTask: Spilling map output
2026-05-12 10:01:54,968 INFO mapred.MapTask: bufstart = 0; bufend = 11983258; bufvoid = 104857600
2026-05-12 10:01:54,968 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22790612(91162448); length = 3423785/6553600
2026-05-12 10:01:55,200 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:01:55,205 INFO mapred.Task: Task:attempt local789236497_0001_m_000007_0 is done. And is in the process of committing
2026-05-12 10:01:55,207 INFO mapred.LocalJobRunner: map
2026-05-12 10:01:55,208 INFO mapred.Task: Task 'attempt local789236497_0001_m_000007_0' done.
2026-05-12 10:01:55,208 INFO mapred.Task: Final Counters for attempt_local789236497_0001_m_000007_0: Counters: 23
File System Counters
  FILE: Number of bytes read=14610
  FILE: Number of bytes written=110185471
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=1073774592
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=19
  HDFS: Number of large read operations=0

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=857977
  Map output records=855947
  Map output bytes=11983258
  Map output materialized bytes=13695158
  Input split bytes=116
  Combine input records=0
  Spilled Records=855947
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=13
  Total committed heap usage (bytes)=637534208
File Input Format Counters
  Bytes Read=134221824
2026-05-12 10:01:55,208 INFO mapred.LocalJobRunner: Finishing task: attempt local789236497_0001_m_000007_0
2026-05-12 10:01:55,208 INFO mapred.LocalJobRunner: Starting task: attempt local789236497_0001_m_000008_0
2026-05-12 10:01:55,209 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:01:55,209 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:01:55,209 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:01:55,210 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:01:55,212 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:1073741824+134217728
2026-05-12 10:01:55,228 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:01:55,228 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:01:55,228 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:01:55,228 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:01:55,228 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:01:55,229 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:01:57,813 INFO mapred.LocalJobRunner:
2026-05-12 10:01:57,814 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:01:57,814 INFO mapred.MapTask: Spilling map output
2026-05-12 10:01:57,814 INFO mapred.MapTask: bufstart = 0; bufend = 11941328; bufvoid = 104857600
2026-05-12 10:01:57,814 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22802592(91210368); length = 3411805/6553600
2026-05-12 10:01:58,052 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:01:58,057 INFO mapred.Task: Task:attempt local789236497_0001_m_000008_0 is done. And is in the process of committing
2026-05-12 10:01:58,059 INFO mapred.LocalJobRunner: map
2026-05-12 10:01:58,059 INFO mapred.Task: Task 'attempt local789236497_0001_m_000008_0' done.
2026-05-12 10:01:58,060 INFO mapred.Task: Final Counters for attempt_local789236497_0001_m_000008_0: Counters: 23
File System Counters
  FILE: Number of bytes read=15633
  FILE: Number of bytes written=123832741
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=1207996416
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=21
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=1
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
Map-Reduce Framework
  Map input records=855198
  Map output records=852952
  Map output bytes=11941328
  Map output materialized bytes=13647238
  Input split bytes=116
  Combine input records=0
  Spilled Records=852952
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=11
  Total committed heap usage (bytes)=637534208
File Input Format Counters
  Bytes Read=134221824
2026-05-12 10:01:58,060 INFO mapred.LocalJobRunner: Finishing task: attempt local789236497_0001_m_000008_0
2026-05-12 10:01:58,060 INFO mapred.LocalJobRunner: Starting task: attempt local789236497_0001_m_000009_0
2026-05-12 10:01:58,061 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:01:58,061 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:01:58,061 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:01:58,062 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:01:58,063 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:1207959552+134217728
2026-05-12 10:01:58,082 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:01:58,082 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:01:58,082 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:01:58,082 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:01:58,082 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:01:58,083 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:02:00,690 INFO mapred.LocalJobRunner:
2026-05-12 10:02:00,691 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:02:00,691 INFO mapred.MapTask: Spilling map output
2026-05-12 10:02:00,691 INFO mapred.MapTask: bufstart = 0; bufend = 11971134; bufvoid = 104857600
2026-05-12 10:02:00,691 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22794076(91176304); length = 3420321/6553600
2026-05-12 10:02:00,925 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:02:00,930 INFO mapred.Task: Task:attempt local789236497_0001_m_000009_0 is done. And is in the process of committing
2026-05-12 10:02:00,933 INFO mapred.LocalJobRunner: map
2026-05-12 10:02:00,933 INFO mapred.Task: Task 'attempt local789236497_0001_m_000009_0' done.
2026-05-12 10:02:00,933 INFO mapred.Task: Final Counters for attempt_local789236497_0001_m_000009_0: Counters: 23
File System Counters
  FILE: Number of bytes read=16144
  FILE: Number of bytes written=137514075
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=1342218240
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=23
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=1
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=857251
  Map output records=855081
  Map output bytes=11971134

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
    Map output bytes=11971134
    Map output materialized bytes=13681302
    Input split bytes=116
    Combine input records=0
    Spilled Records=855001
    Failed Shuffles=0
    Merged Map outputs=0
    GC time elapsed (ms)=11
    Total committed heap usage (bytes)=637534208
File Input Format Counters
    Bytes Read=134221824
2026-05-12 10:02:00,933 INFO mapred.LocalJobRunner: Finishing task: attempt local789236497_0001_m_000009_0
2026-05-12 10:02:00,933 INFO mapred.LocalJobRunner: Starting task: attempt local789236497_0001_m_000010_0
2026-05-12 10:02:00,934 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:02:00,935 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:02:00,935 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:02:00,935 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:02:00,937 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:1342177280+134217728
2026-05-12 10:02:00,956 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:02:00,956 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:02:00,956 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:02:00,956 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:02:00,956 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:02:00,957 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:02:03,595 INFO mapred.LocalJobRunner:
2026-05-12 10:02:03,595 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:02:03,595 INFO mapred.MapTask: Spilling map output
2026-05-12 10:02:03,595 INFO mapred.MapTask: bufstart = 0; bufend = 11959696; bufvoid = 104857600
2026-05-12 10:02:03,595 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22797344(91189376); length = 3417053/6553600
2026-05-12 10:02:03,829 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:02:03,835 INFO mapred.Task: Task:attempt local789236497_0001_m_000010_0 is done. And is in the process of committing
2026-05-12 10:02:03,837 INFO mapred.LocalJobRunner: map
2026-05-12 10:02:03,838 INFO mapred.Task: Task 'attempt local789236497_0001_m_000010_0' done.
2026-05-12 10:02:03,838 INFO mapred.Task: Final Counters for attempt local789236497_0001_m_000010_0: Counters: 23
File System Counters
    FILE: Number of bytes read=16655
    FILE: Number of bytes written=151182337
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=1476440064
    HDFS: Number of bytes written=0
    HDFS: Number of read operations=25
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=1
    HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
    Map input records=856497
    Map output records=854264
    Map output bytes=11959696
    Map output materialized bytes=13668230
    Input split bytes=116
    Combine input records=0

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
    Combine input records=0
    Spilled Records=854264
    Failed Shuffles=0
    Merged Map outputs=0
    GC time elapsed (ms)=15
    Total committed heap usage (bytes)=637534208
File Input Format Counters
    Bytes Read=134221824
2026-05-12 10:02:03,838 INFO mapred.LocalJobRunner: Finishing task: attempt local789236497_0001_m_000010_0
2026-05-12 10:02:03,838 INFO mapred.LocalJobRunner: Starting task: attempt local789236497_0001_m_000011_0
2026-05-12 10:02:03,839 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:02:03,839 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:02:03,839 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:02:03,840 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:02:03,842 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:1476395008+134217728
2026-05-12 10:02:03,860 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:02:03,860 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:02:03,860 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:02:03,860 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:02:03,860 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:02:03,861 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:02:06,480 INFO mapred.LocalJobRunner:
2026-05-12 10:02:06,481 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:02:06,481 INFO mapred.MapTask: Spilling map output
2026-05-12 10:02:06,481 INFO mapred.MapTask: bufstart = 0; bufend = 11934790; bufvoid = 104857600
2026-05-12 10:02:06,481 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 22804460(91217840); length = 3409937/6553600
2026-05-12 10:02:06,709 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:02:06,714 INFO mapred.Task: Task:attempt local789236497_0001_m_000011_0 is done. And is in the process of committing
2026-05-12 10:02:06,717 INFO mapred.LocalJobRunner: map
2026-05-12 10:02:06,717 INFO mapred.Task: Task 'attempt local789236497_0001_m_000011_0' done.
2026-05-12 10:02:06,717 INFO mapred.Task: Final Counters for attempt local789236497_0001_m_000011_0: Counters: 23
File System Counters
    FILE: Number of bytes read=17166
    FILE: Number of bytes written=164822135
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=1610661888
    HDFS: Number of bytes written=0
    HDFS: Number of read operations=27
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=1
    HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
    Map input records=854684
    Map output records=852485
    Map output bytes=11934790
    Map output materialized bytes=13639766
    Input split bytes=116
    Combine input records=0
    Spilled Records=852485
    Failed Shuffles=0
    Merged Map outputs=0

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
Merged Map outputs=0
GC time elapsed (ms)=19
Total committed heap usage (bytes)=637534208
File Input Format Counters
  Bytes Read=134221824
2026-05-12 10:02:06,717 INFO mapred.LocalJobRunner: Finishing task: attempt local789236497_0001_m_000011_0
2026-05-12 10:02:06,717 INFO mapred.LocalJobRunner: Starting task: attempt local789236497_0001_m_000012_0
2026-05-12 10:02:06,718 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:02:06,718 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:02:06,718 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:02:06,718 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:02:06,719 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/real_input/real_taxi_data.csv:1610612736+98061756
2026-05-12 10:02:06,735 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-05-12 10:02:06,735 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-05-12 10:02:06,735 INFO mapred.MapTask: soft limit at 83886080
2026-05-12 10:02:06,735 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-05-12 10:02:06,735 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-05-12 10:02:06,736 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-05-12 10:02:08,683 INFO mapred.LocalJobRunner:
2026-05-12 10:02:08,683 INFO mapred.MapTask: Starting flush of map output
2026-05-12 10:02:08,683 INFO mapred.MapTask: Spilling map output
2026-05-12 10:02:08,683 INFO mapred.MapTask: bufstart = 0; bufend = 8715462; bufvoid = 104857600
2026-05-12 10:02:08,683 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 23724268(94897072); length = 2490129/6553600
2026-05-12 10:02:08,852 INFO mapred.MapTask: Finished spill 0
2026-05-12 10:02:08,857 INFO mapred.Task: Task:attempt local789236497_0001_m_000012_0 is done. And is in the process of committing
2026-05-12 10:02:08,860 INFO mapred.LocalJobRunner: map
2026-05-12 10:02:08,860 INFO mapred.Task: Task 'attempt local789236497_0001_m_000012_0' done.
2026-05-12 10:02:08,860 INFO mapred.Task: Final Counters for attempt local789236497_0001_m_000012_0: Counters: 23
File System Counters
  FILE: Number of bytes read=17677
  FILE: Number of bytes written=174782701
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=1708723644
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=29
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=1
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=624281
  Map output records=622533
  Map output bytes=8715462
  Map output materialized bytes=9960534
  Input split bytes=116
  Combine input records=0
  Spilled Records=622533
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=8
  Total committed heap usage (bytes)=637534208
File Input Format Counters

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
File Input Format Counters
  Bytes Read=98061756
2026-05-12 10:02:08,860 INFO mapred.LocalJobRunner: Finishing task: attempt local789236497_0001_m_000012_0
2026-05-12 10:02:08,860 INFO mapred.LocalJobRunner: map task executor complete.
2026-05-12 10:02:08,863 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-05-12 10:02:08,863 INFO mapred.LocalJobRunner: Starting task: attempt local789236497_0001_r_000000_0
2026-05-12 10:02:08,874 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-12 10:02:08,874 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-12 10:02:08,874 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup
failures: false
2026-05-12 10:02:08,874 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-05-12 10:02:08,878 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle@46326155
2026-05-12 10:02:08,880 WARN Impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-05-12 10:02:08,895 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit=2814168320, maxSingleShuffleLimit=703542080, mergeThreshold=1857351168, i
oSortFactor=10, memToMemMergeOutputsThreshold=10
2026-05-12 10:02:08,897 INFO reduce.EventFetcher: attempt local789236497_0001_r_000000_0 Thread started: EventFetcher for fetching Map Completion Events
2026-05-12 10:02:08,925 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt local789236497_0001_m_000006_0 decomp: 13666098 le
n: 13666102 to MEMORY
2026-05-12 10:02:08,941 INFO reduce.InMemoryMapOutput: Read 13666098 bytes from map-output for attempt local789236497_0001_m_000006_0
2026-05-12 10:02:08,943 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 13666098, inMemoryMapOutputs.size() -> 1, commitMemory -> 0
, usedMemory ->13666098
2026-05-12 10:02:08,946 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt local789236497_0001_m_000012_0 decomp: 9960530 len
: 9960534 to MEMORY
2026-05-12 10:02:08,956 INFO reduce.InMemoryMapOutput: Read 9960530 bytes from map-output for attempt local789236497_0001_m_000012_0
2026-05-12 10:02:08,957 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 9960530, inMemoryMapOutputs.size() -> 2, commitMemory -> 13
666098, usedMemory ->23626628
2026-05-12 10:02:08,960 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt local789236497_0001_m_000007_0 decomp: 13695154 le
n: 13695158 to MEMORY
2026-05-12 10:02:08,973 INFO reduce.InMemoryMapOutput: Read 13695154 bytes from map-output for attempt local789236497_0001_m_000007_0
2026-05-12 10:02:08,973 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 13695154, inMemoryMapOutputs.size() -> 3, commitMemory -> 2
3626628, usedMemory ->37321782
2026-05-12 10:02:08,976 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt local789236497_0001_m_000000_0 decomp: 13680450 le
n: 13680454 to MEMORY
2026-05-12 10:02:08,985 INFO reduce.InMemoryMapOutput: Read 13680450 bytes from map-output for attempt local789236497_0001_m_000000_0
2026-05-12 10:02:08,985 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 13680450, inMemoryMapOutputs.size() -> 4, commitMemory -> 3
7321782, usedMemory ->51002232
2026-05-12 10:02:08,987 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt local789236497_0001_m_000001_0 decomp: 13757410 le
n: 13757414 to MEMORY
2026-05-12 10:02:09,001 INFO reduce.InMemoryMapOutput: Read 13757410 bytes from map-output for attempt local789236497_0001_m_000001_0
2026-05-12 10:02:09,001 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 13757410, inMemoryMapOutputs.size() -> 5, commitMemory -> 5
1002232, usedMemory ->64759642
2026-05-12 10:02:09,005 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt local789236497_0001_m_000002_0 decomp: 13652194 le
n: 13652198 to MEMORY
2026-05-12 10:02:09,013 INFO reduce.InMemoryMapOutput: Read 13652194 bytes from map-output for attempt local789236497_0001_m_000002_0
2026-05-12 10:02:09,013 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 13652194, inMemoryMapOutputs.size() -> 6, commitMemory -> 6
4759642, usedMemory ->78411836
2026-05-12 10:02:09,016 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt local789236497_0001_m_000008_0 decomp: 13647234 le
n: 13647238 to MEMORY
2026-05-12 10:02:09,023 INFO reduce.InMemoryMapOutput: Read 13647234 bytes from map-output for attempt local789236497_0001_m_000008_0
2026-05-12 10:02:09,024 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 13647234, inMemoryMapOutputs.size() -> 7, commitMemory -> 7
8411836, usedMemory ->92059070
2026-05-12 10:02:09,027 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt local789236497_0001_m_000003_0 decomp: 13695906 le
n: 13695910 to MEMORY
2026-05-12 10:02:09,034 INFO reduce.InMemoryMapOutput: Read 13695906 bytes from map-output for attempt local789236497_0001_m_000003_0

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
2026-05-12 10:02:09,070 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt_local789236497_0001_m_000004_0 decomp: 13664002 le
n: 13664006 to MEMORY
2026-05-12 10:02:09,079 INFO reduce.InMemoryMapOutput: Read 13664002 bytes from map-output for attempt_local789236497_0001_m_000004_0
2026-05-12 10:02:09,079 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 13664002, inMemoryMapOutputs.size() -> 12, commitMemory ->
146744262, usedMemory ->160408264
2026-05-12 10:02:09,082 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt_local789236497_0001_m_000005_0 decomp: 13663474 le
n: 13663478 to MEMORY
2026-05-12 10:02:09,090 INFO reduce.InMemoryMapOutput: Read 13663474 bytes from map-output for attempt_local789236497_0001_m_000005_0
2026-05-12 10:02:09,091 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 13663474, inMemoryMapOutputs.size() -> 13, commitMemory ->
160408264, usedMemory ->174071738
2026-05-12 10:02:09,091 INFO reduce.EventFetcher: EventFetcher is interrupted.. Returning
2026-05-12 10:02:09,092 INFO mapred.LocalJobRunner: 13 / 13 copied.
2026-05-12 10:02:09,092 INFO reduce.MergeManagerImpl: finalMerge called with 13 in-memory map-outputs and 0 on-disk map-outputs
2026-05-12 10:02:09,101 INFO mapred.Merger: Merging 13 sorted segments
2026-05-12 10:02:09,101 INFO mapred.Merger: Down to the last merge-pass, with 13 segments left of total size: 174071634 bytes
2026-05-12 10:02:11,745 INFO reduce.MergeManagerImpl: Merged 13 segments, 174071738 bytes to disk to satisfy reduce memory limit
2026-05-12 10:02:11,746 INFO reduce.MergeManagerImpl: Merging 1 files, 174071718 bytes from disk
2026-05-12 10:02:11,746 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from memory into reduce
2026-05-12 10:02:11,746 INFO mapred.Merger: Merging 1 sorted segments
2026-05-12 10:02:11,747 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 174071706 bytes
2026-05-12 10:02:11,747 INFO mapred.LocalJobRunner: 13 / 13 copied.
2026-05-12 10:02:11,770 INFO Configuration.deprecation: mapred.skip.on is deprecated. Instead, use mapreduce.job.skiprecords
2026-05-12 10:02:14,596 INFO mapred.Task: Task:attempt_local789236497_0001_r_000000_0 is done. And is in the process of committing
2026-05-12 10:02:14,599 INFO mapred.LocalJobRunner: 13 / 13 copied.
2026-05-12 10:02:14,599 INFO mapred.Task: Task attempt_local789236497_0001_r_000000_0 is allowed to commit now
2026-05-12 10:02:14,612 INFO output.FileOutputCommitter: Saved output of task 'attempt_local789236497_0001_r_000000_0' to hdfs://localhost:9000/real_outpu
t
2026-05-12 10:02:14,613 INFO mapred.LocalJobRunner: reduce > reduce
2026-05-12 10:02:14,613 INFO mapred.Task: Task 'attempt_local789236497_0001_r_000000_0' done.
2026-05-12 10:02:14,613 INFO mapred.Task: Final Counters for attempt_local789236497_0001_r_000000_0: Counters: 30
File System Counters
  FILE: Number of bytes read=348161601
  FILE: Number of bytes written=348854419
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=1708723644
  HDFS: Number of bytes written=285
  HDFS: Number of read operations=34
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=3
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Combine input records=0
  Combine output records=0
  Reduce input groups=24
  Reduce shuffle bytes=174071790
  Reduce input records=10879482
  Reduce output records=24
  Spilled Records=10879482
  Shuffled Maps =13
  Failed Shuffles=0
  Merged Map outputs=13
  GC time elapsed (ms)=3

```

```

pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hadoop jar gridlock.jar GridlockDecoder /real input /real output
  GC time elapsed (ms)=3
  Total committed heap usage (bytes)=637534208
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Output Format Counters
  Bytes Written=285
2026-05-12 10:02:14,613 INFO mapred.LocalJobRunner: Finishing task: attempt_local789236497_0001_r_000000_0
2026-05-12 10:02:14,614 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-05-12 10:02:15,234 INFO mapreduce.Job: map 100% reduce 100%
2026-05-12 10:02:15,235 INFO mapreduce.Job: Job job_local789236497_0001 completed successfully
2026-05-12 10:02:15,253 INFO mapreduce.Job: Counters: 36
File System Counters
  FILE: Number of bytes read=348327992
  FILE: Number of bytes written=1599485608
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=13886749560
  HDFS: Number of bytes written=285
  HDFS: Number of read operations=255
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=16
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=10906859
  Map output records=10879482
  Map output bytes=152312748
  Map output materialized bytes=174071790
  Input split bytes=1508
  Combine input records=0
  Combine output records=0
  Reduce input groups=24
  Reduce shuffle bytes=174071790
  Reduce input records=10879482
  Reduce output records=24
  Spilled Records=21758964
  Shuffled Maps =13
  Failed Shuffles=0
  Merged Map outputs=13
  GC time elapsed (ms)=258
  Total committed heap usage (bytes)=8387559424
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters

```

After this is done, we can see the results, the output.

```
File Input Format Counters
  Bytes Read=1708723644
File Output Format Counters
  Bytes Written=285
pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ hdfs dfs -cat /real output/part-r-00000
2026-05-12 10:04:15,285 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
00:00 12.5
01:00 12.08
02:00 11.85
03:00 11.93
04:00 12.4
05:00 11.97
06:00 11.35
07:00 12.33
08:00 13.52
09:00 13.45
10:00 13.42
11:00 13.45
12:00 13.39
13:00 13.62
14:00 14.44
15:00 14.72
16:00 14.6
17:00 14.49
18:00 13.52
19:00 12.56
20:00 12.37
21:00 12.38
22:00 12.63
23:00 12.72
pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$ stop-all.sh
WARNING: Stopping all Apache Hadoop daemons as pc in 10 seconds.
WARNING: Use CTRL-C to abort.
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [pc]
2026-05-12 10:05:42,314 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Stopping nodemanagers
Stopping resourcemanager
pc@pc:~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder$
```

We can observe a list with all 24 hours of the day and the avg. time for a trip.

Some screenshots of code:

```
public class GridlockDecoder {

    public static class TaxiMapper extends Mapper<Object, Text, Text, DoubleWritable> {
        private Text hourKey = new Text();
        private DoubleWritable durationValue = new DoubleWritable();
        private SimpleDateFormat format = new SimpleDateFormat("yyyy-MM-dd HH:mm:ss");
        private SimpleDateFormat hourFormat = new SimpleDateFormat("HH");

        public void map(Object key, Text value, Context context) throws IOException, InterruptedException {
            String line = value.toString();
            if (line.contains("VendorID")) return; // Skip the header row

            String[] columns = line.split(",");
            if (columns.length > 2) {
                try {
                    Date pickup = format.parse(columns[1]);
                    Date dropoff = format.parse(columns[2]);

                    // Extract hour (e.g., "08")
                    String hour = hourFormat.format(pickup);

                    // Calculate duration in minutes
                    double minutes = (dropoff.getTime() - pickup.getTime()) / 60000.0;

                    if (minutes > 0 && minutes < 600) {
                        hourKey.set(hour + ":00"); // Format as "08:00"
                        durationValue.set(minutes);
                        context.write(hourKey, durationValue); // Send to Reducer
                    }
                } catch (Exception e) { /* Ignore bad data rows */ }
            }
        }
    }
}
```

```

public static class AverageReducer extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {
    private DoubleWritable result = new DoubleWritable();

    public void reduce(Text key, Iterable<DoubleWritable> values, Context context) throws IOException, InterruptedException {
        double sum = 0;
        int count = 0;
        for (DoubleWritable val : values) {
            sum += val.get();
            count++;
        }
        if (count > 0) {
            double avg = Math.round((sum / count) * 100.0) / 100.0; // Round to 2 decimals
            result.set(avg);
            context.write(key, result); // Final output
        }
    }
}

```

```

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Gridlock Decoder");
    job.setJarByClass(GridlockDecoder.class);
    job.setMapperClass(TaxiMapper.class);
    job.setReducerClass(AverageReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(DoubleWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}

```

How to run the project:

GitHub Link: <https://github.com/EdwardAdelin/Grid-Lock-Decoder>

Use the “git clone” command to copy the repo to your local machine.

Follow these instructions that can be also found in the ReadMe file.

Navigate to folder

```
cd ~/Music/DPA-Project-A4-S2/Grid-Lock-Decoder
```

Start Hadoop

In terminal:

```
start-dfs.sh
```

```
start-yarn.sh
```

and then verify it is running

```
jps
```

(Make sure you see the 5 processes: NameNode, DataNode, SecondaryNameNode, ResourceManager, NodeManager).

Reset (!!! CRITICAL !!!)

Hadoop has a strict safety feature: It will instantly crash if the output folder already exists. Since you already ran this project successfully, the /real_output folder is currently sitting in HDFS.

Right before you start the demo, delete the old output folder so your live run has a clean slate:

```
hdfs dfs -rm -r /real_output
```

If that gives error then:

```
stop-all.sh
```

and then

```
hdfs namenode -format
```

Then start from begining.

Continuation

1. Create the input folder

```
hdfs dfs -mkdir -p /real_input
```

2. Upload the CSV file (This will take a minute)

```
hdfs dfs -put real_taxi_data.csv /real_input/
```

Ready for Demo Phase

You are completely back to normal. Because the drive was wiped, you do not need to run the `rm -r /real_output` reset command.

```
hadoop jar gridlock.jar GridlockDecoder /real_input /real_output
```

Because we are running this on a single-node Virtual Machine, Ubuntu clears the /tmp directory on reboot, which drops the HDFS volumes. In a real production environment, we would configure the `hdfs-site.xml` to store the datanode blocks on a permanent, dedicated physical drive so data persists between server reboots

Use the `cat` command to print the contents of the final output file to your terminal.

Showing Results

```
hdfs dfs -cat /real_output/part-r-00000
```

Then we can stop with

```
stop-all.sh
```

Installation, implementation, testing

1. Introduction and Objectives

The objective of this project is to implement a Distributed Parallel Algorithm using Apache Hadoop to process and analyze a massive real-world dataset. The project, titled "Gridlock Decoder," analyzes the NYC Yellow Taxi Trip dataset (January 2016) containing over 10 million rows. By utilizing the MapReduce programming model, the algorithm calculates the average trip duration for every hour of the day to identify daily traffic bottlenecks and optimal travel times in New York City.

2. Environment Setup and Installation

2.1. Virtual Machine Configuration

Following the laboratory guidelines, the environment was built using a virtualized Linux distribution.

1. Installed **Oracle VM VirtualBox**.
2. Created a new Virtual Machine running **Ubuntu (64-bit)**. Virtualization was enabled in the host machine's BIOS (Intel VT-x) to support the 64-bit architecture.

2.2. Hadoop Installation and Configuration

Apache Hadoop was installed and configured to run in a Single-Node Pseudo-Distributed mode.

1. Installed Java Development Kit (JDK) and OpenSSH server.
2. Generated passwordless SSH keys to allow Hadoop daemons to communicate locally without prompts:
3. Bash

```
ssh-keygen -t rsa -P "" -f ~/.ssh/id_rsa
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
chmod 0600 ~/.ssh/authorized_keys
```

4. Configured the Hadoop environment variables in `~/.bashrc`:
5. Bash

```
export HADOOP_HOME=/home/pc/hadoop
```

```
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
```

6. Modified the core Hadoop configuration files:
 - **core-site.xml**: Defined the default filesystem URI to point to the local network port (hdfs://localhost:9000).
 - **hdfs-site.xml**: Set the dfs.replication property to 1, as this is a single-node cluster.
7. Formatted the Hadoop Distributed File System (HDFS) NameNode:
8. Bash

```
hdfs namenode -format
```

3. Data Acquisition and Preparation

The raw dataset was sourced from Kaggle (NYC Yellow Taxi Trip Data, Jan 2016). The compressed .zip file was downloaded to the Ubuntu machine, extracted, and renamed for easier access.

To allow the MapReduce algorithm to process the data, it was uploaded from the local Linux filesystem into HDFS:

Bash

```
# Create the input directory in HDFS
```

```
hdfs dfs -mkdir -p /real_input
```

```
# Upload the 1.5GB CSV file to HDFS
```

```
hdfs dfs -put real_taxi_data.csv /real_input/
```

4. Implementation

The project relies on a custom Java program that defines the MapReduce jobs.

4.1. The Algorithm Design

- **The Mapper (TaxiMapper)**: Reads the CSV file line by line. It skips the header row. For valid rows, it extracts the pickup timestamp, parses the specific "Hour of the Day" (e.g., "08:00"), and calculates the total trip duration in minutes using the difference between drop-off and pickup times. It outputs Key-Value pairs: [Hour, Duration].
- **The Shuffle and Sort Phase**: Hadoop automatically routes all output values with the exact same Hour key to the same Reducer.
- **The Reducer (AverageReducer)**: Receives an iterable list of all trip durations for a specific hour. It sums these values, divides by the total count to find the average, and writes the final Key-Value pair to HDFS.
- *Note on Combiners*: A Combiner was not used in this project. Combiners require operations to be associative and commutative (like summation). Because this

algorithm calculates an average, applying a Combiner locally would mathematically corrupt the final global averages.

4.2. Java Source Code (GridlockDecoder . java)

Java

```
import java.io.IOException;
import java.text.SimpleDateFormat;
import java.util.Date;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class GridlockDecoder {

    public static class TaxiMapper extends Mapper<Object, Text, Text, DoubleWritable> {
        private Text hourKey = new Text();
        private DoubleWritable durationValue = new DoubleWritable();
        private SimpleDateFormat format = new SimpleDateFormat("yyyy-MM-dd HH:mm:ss");
        private SimpleDateFormat hourFormat = new SimpleDateFormat("HH");

        public void map(Object key, Text value, Context context) throws IOException,
        InterruptedException {
            String line = value.toString();
            if (line.contains("VendorID")) return;

            String[] columns = line.split(",");
            if (columns.length > 2) {
                try {
                    Date pickup = format.parse(columns[1]);
                    Date dropoff = format.parse(columns[2]);

                    String hour = hourFormat.format(pickup);
                    double minutes = (dropoff.getTime() - pickup.getTime()) / 60000.0;

                    if (minutes > 0 && minutes < 600) {
                        hourKey.set(hour + ":00");
                        durationValue.set(minutes);
                        context.write(hourKey, durationValue);
                    }
                } catch (Exception e) {}
            }
        }
    }
}
```

```

    }
}

public static class AverageReducer extends Reducer<Text, DoubleWritable, Text,
DoubleWritable> {
    private DoubleWritable result = new DoubleWritable();

    public void reduce(Text key, Iterable<DoubleWritable> values, Context context) throws
IOException, InterruptedException {
        double sum = 0;
        int count = 0;
        for (DoubleWritable val : values) {
            sum += val.get();
            count++;
        }
        if (count > 0) {
            double avg = Math.round((sum / count) * 100.0) / 100.0;
            result.set(avg);
            context.write(key, result);
        }
    }
}

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Gridlock Decoder");
    job.setJarByClass(GridlockDecoder.class);
    job.setMapperClass(TaxiMapper.class);
    job.setReducerClass(AverageReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(DoubleWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

5. Execution and Testing

5.1. Compilation and Packaging

The Java code was compiled using Hadoop's libraries and packaged into a JAR file for cluster distribution:

Bash

```
hadoop com.sun.tools.javac.Main GridlockDecoder.java
```

```
jar cf gridlock.jar GridlockDecoder*.class
```

5.2. Running the MapReduce Job

The Hadoop cluster daemons were started (`start-dfs.sh`, `start-yarn.sh`), verified using the `jps` command, and the job was submitted to YARN:

```
Bash
hadoop jar gridlock.jar GridlockDecoder /real_input /real_output
```

5.3. Process Monitoring and Parallelism Verification

During execution, the Hadoop console output demonstrated successful parallelization. The framework identified the 1.5GB file size and mathematically divided the input dataset into 13 separate chunks, triggering 13 parallel Map tasks: `INFO mapreduce.JobSubmitter: number of splits: 13`

The counters verified that all records were processed and aggregated into exactly 24 output groups (representing the 24 hours of the day):

```
Plaintext
Map input records=10906859
Map output records=10879482
Reduce input groups=24
Reduce output records=24
```

5.4. Results

The final results were read directly from HDFS:

```
Bash
hdfs dfs -cat /real_output/part-r-00000
```

Output:

```
Plaintext
00:00 12.5
01:00 12.08
...
06:00 11.35
...
15:00 14.72
...
23:00 12.72
```

6. Conclusion

The implementation of the MapReduce algorithm successfully processed nearly 11 million rows of taxi telemetry. The output explicitly maps out urban traffic bottlenecks in New York City. The data reveals that the most optimal time for travel is 6:00 AM (averaging 11.35 minutes per trip), while peak gridlock occurs at 3:00 PM (15:00), pushing the average trip duration up to 14.72 minutes. The project demonstrates the efficiency of Hadoop in horizontally scaling data processing tasks that would otherwise overwhelm traditional sequential applications.

Code

Visit: <https://github.com/EdwardAdelin/Grid-Lock-Decoder>

GitHub has the ready to use Jar file.

GridLock Java:

```
import java.io.IOException;
import java.text.SimpleDateFormat;
import java.util.Date;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class GridlockDecoder {

    public static class TaxiMapper extends Mapper<Object, Text, Text, DoubleWritable> {
        private Text hourKey = new Text();
        private DoubleWritable durationValue = new DoubleWritable();
        private SimpleDateFormat format = new SimpleDateFormat("yyyy-MM-dd HH:mm:ss");
        private SimpleDateFormat hourFormat = new SimpleDateFormat("HH");

        public void map(Object key, Text value, Context context) throws IOException, InterruptedException {
            String line = value.toString();
            if (line.contains("VendorID")) return;

            String[] columns = line.split(",");
            if (columns.length > 2) {
                try {
                    Date pickup = format.parse(columns[1]);
                    Date dropoff = format.parse(columns[2]);

                    String hour = hourFormat.format(pickup);
                    double minutes = (dropoff.getTime() - pickup.getTime()) / 60000.0;

                    if (minutes > 0 && minutes < 600) {
                        hourKey.set(hour + ":00");
                        durationValue.set(minutes);
                        context.write(hourKey, durationValue);
                    }
                } catch (Exception e) {}
            }
        }
    }
}
```

```

public static class AverageReducer extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {
    private DoubleWritable result = new DoubleWritable();

    public void reduce(Text key, Iterable<DoubleWritable> values, Context context) throws IOException, InterruptedException {
        double sum = 0;
        int count = 0;
        for (DoubleWritable val : values) {
            sum += val.get();
            count++;
        }
        if (count > 0) {
            double avg = Math.round((sum / count) * 100.0) / 100.0;
            result.set(avg);
            context.write(key, result);
        }
    }
}

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Gridlock Decoder");
    job.setJarByClass(GridlockDecoder.class);
    job.setMapperClass(TaxiMapper.class);
    job.setReducerClass(AverageReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(DoubleWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

Bash code used:

1. Start the Hadoop Cluster

start-dfs.sh

start-yarn.sh

2. Upload the Kaggle Dataset to HDFS

hdfs dfs -mkdir -p /real_input

hdfs dfs -put real_taxi_data.csv /real_input/

3. Compile and Package the Java Code (If re-compiling is necessary)

hadoop com.sun.tools.javac.Main GridlockDecoder.java

jar cf gridlock.jar GridlockDecoder*.class

4. Execute the MapReduce Algorithm

hadoop jar gridlock.jar GridlockDecoder /real_input /real_output

5. View the Final Analytics

hdfs dfs -cat /real_output/part-r-00000